



English, Economics, Politics, and Philosophy

Unit 3 Cases in Business

Lesson 3.3: Elon Musk and Tesla Motors SpaceX

Pertinent Topics:

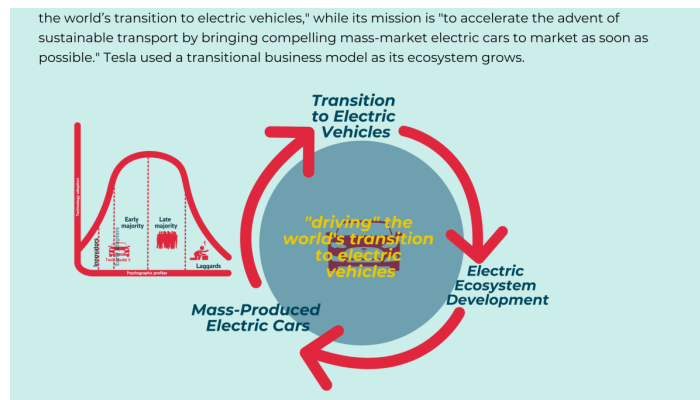
- Elon Musk
- Business life cycle
- Tesla Motors

Success Criteria:

- I will learn about the story behind Musk's succeed
- I will be able to identify Musk's managerial style
- I will learn the business life cycle of Tesla

MINDS ON

Imagine a world where gasoline-fueled automobiles are eliminated. What may be some of the pros as a result? What would be some of the cons?



Pro: The pollution problem of our environment may be mitigated/ improved.

Con: The usage of other kind of fuel may still be harmful to our environment.

I. Elon Musk (1971-)

Musk was born in Pretoria, South Africa, to Maye (née Haldeman), a model, and Errol Musk, a businessman and engineer. Musk briefly attended the University of Pretoria before immigrating to Canada at the age of 18, acquiring citizenship through his



Canadian-born mother. Two years later he matriculated at **Queen's University at Kingston in Canada**. Musk later transferred to the University of Pennsylvania and received bachelor's degrees in **economics and physics**.

He moved to California in 1995 to attend **Stanford University**, but dropped out after two days and, with his brother Kimbal, co-founded the online city guide software company Zip2.

After co-founding and selling Zip2 and **PayPal**, Musk amassed significant capital and honed his skills in technology innovation and business scaling. His experience with **SpaceX, founded in 2002 using \$100 million from the PayPal sale, demonstrated his ability to tackle complex engineering challenges and think long-term, showcasing his vision and risk-taking abilities**.

In 2004, Musk became an early investor in Tesla Motors, providing most of the initial financing and becoming the company's chairman. His hands-on leadership as product architect and later CEO in 2008 was pivotal in steering Tesla toward groundbreaking developments in electric vehicles and sustainable energy solutions. Musk's prior experiences equipped him with the technical expertise, innovative mindset, and bold leadership style that transformed Tesla into a leading force in the automotive industry, advancing technologies like autonomous driving and integrating solar energy through the acquisition of SolarCity.

Q1: Is there any commonality shared by Altman and Musk?

website for B2B services Business to Business

Now, let's have a brief glance at his entrepreneurial career thus far.

- Online Banking system
1. **Early Entrepreneurial Ventures:** Musk's founding roles in **Zip2** and **PayPal** provided him with valuable experience in starting and scaling technology companies. Zip2, an online city guide software, was sold to Compaq for \$307 million, giving Musk significant capital and business acumen. His subsequent venture, **PayPal**, revolutionized online payments and was acquired by eBay for \$1.5 billion. These successes not only increased his wealth but also solidified his reputation as a forward-thinking entrepreneur.
 2. **Risk-Taking and Vision:** Musk's willingness to invest in ambitious projects set the stage for Tesla's innovative approach. After PayPal, he founded **SpaceX** with the goal of reducing space transportation costs. This demonstrated his ability to tackle large-scale challenges and think long-term, qualities that he brought to Tesla when he joined the company in 2004 as chairman and later became CEO.
 3. **Technical Expertise and Innovation:** With a background in physics and economics, Musk possesses a strong understanding of complex technologies. His hands-on involvement in product development at Tesla led to groundbreaking vehicles like the **Model S, Model 3, and Model Y**. His focus on innovation drove the company to develop its own battery technology and invest in

Why would someone be risk averse/lover

1. Maybe some people are more adventurous whereas others much prefer stay in the comfort zone.

building **Gigafactories**, which streamlined production and reduced costs.

4. **Leadership and Management Style:** Musk's leadership is characterized by **high standards and a focus on efficiency**. He encourages a culture of hard work and creativity, pushing teams to achieve ambitious goals. His **direct communication style and ability to inspire** have been instrumental in keeping Tesla agile and **competitive**.
5. **Financial Investment and Resources:** Musk's personal investment in Tesla helped the company survive early financial struggles. His ability to **secure funding and manage resources efficiently** allowed Tesla to invest heavily in research and **development**, manufacturing, and infrastructure.
6. **Brand Building and Marketing:** Musk's public persona and savvy use of social media have generated significant publicity for Tesla at minimal cost. His ability to generate excitement around product launches and company milestones has helped build a strong and loyal customer base.
7. **Strategic Vision for Sustainable Energy:** Musk's commitment to sustainable energy solutions aligned with growing environmental concerns worldwide. By positioning **Tesla not just as a car manufacturer but as an energy company**, he expanded its market and influence. Initiatives like **Tesla Energy and the acquisition of**

R&D: Research and Development

SolarCity integrated renewable energy generation and storage, reinforcing the company's mission.

8. **Vertical Integration and Control Over Supply Chain:** Learning from challenges in previous ventures, Musk emphasized the importance of controlling the supply chain. Tesla's vertical integration, from battery production to software development, has allowed for greater quality control and innovation speed.

II. Product Life Cycle PLC

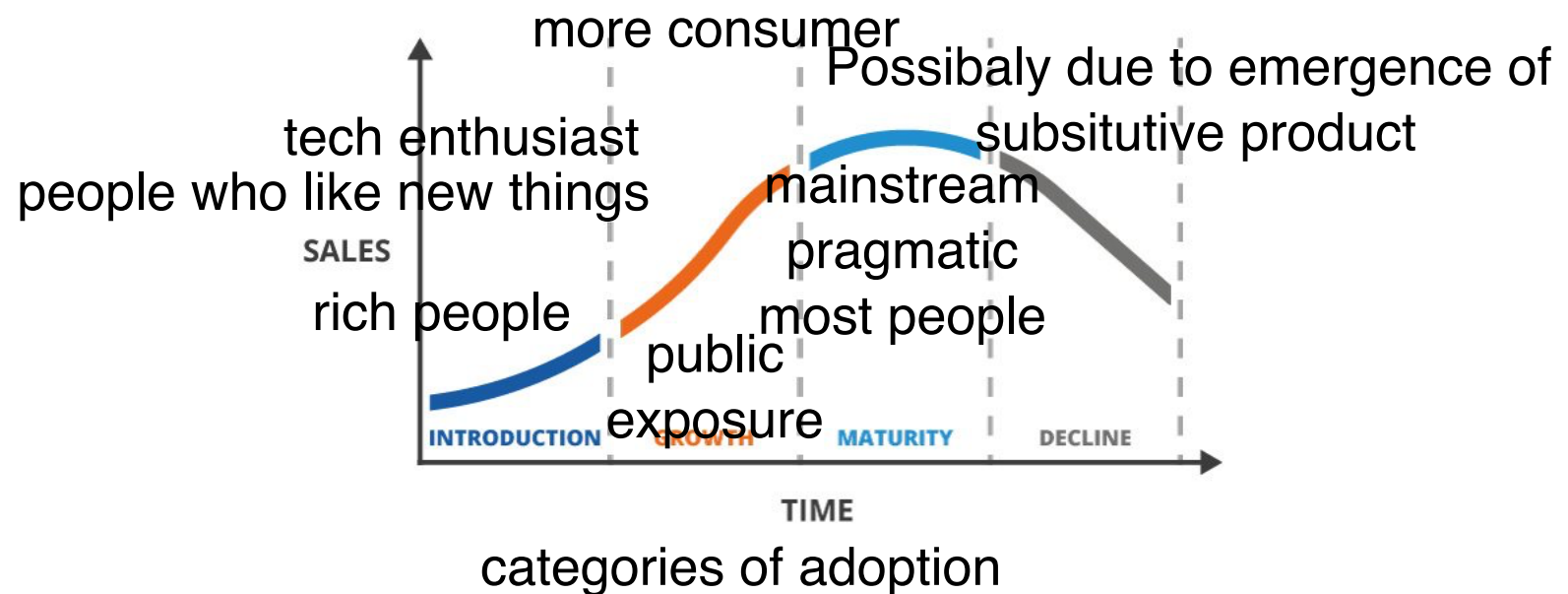
Key Takeaways:

the life of product

1. A product **life cycle** is the length of time from when a product is introduced to the market until it's taken off shelves.
2. There are four stages in a product's life cycle: **introduction, growth, maturity, and decline**.
3. A company often incurs **higher marketing costs** when **introducing a product** to the market but **experiences higher sales as product adoption grows**.
4. Sales stabilize and peak when the product's **adoption matures**, though competition and obsolescence may cause its decline.
5. The concept of product life cycle helps inform marketing and sales decisions, from **pricing and promotion to expansion or cost-cutting**.

economic + statistic, we can derive PLC

PRODUCT LIFE CYCLE



Introduction Stage

The introduction stage is when a new product is **launched**, and customers encounter it for the first time. Companies invest heavily in **advertising to build awareness and educate consumers about the product's benefits, especially if it's a novel offering**. At this point, competition is minimal or nonexistent, but the business often faces negative financial results due to low sales volumes and high marketing expenditures. The focus is on refining the sales strategy and establishing a market presence.

marketing(advertising) to popularity
introduction to growth

Growth Stage

In the growth stage, the product begins to gain **popularity**, leading to increased demand, higher production levels, and expanded availability. Companies may continue substantial marketing efforts to **differentiate their product** from emerging competitors. Enhancements and improvements are often made based on customer feedback. Financially, this phase sees rising sales and revenue, but as competitors enter the market, companies might experience pressure to reduce prices, which can narrow profit margins.

innovate and stand out (to gain a competitive advantage)
at any stage,

it is possible for a product to reach its declining stage

Melissa: when you have too much of something
that something loses its value.

(law of scarcity)

Maturity Stage

The maturity stage is characterized by **peak profitability as production and marketing costs decline**. The market becomes saturated with similar products, intensifying competition and causing **profit margins to shrink**. Sales volumes level off, signaling that the product has maxed out its market potential. **Companies strive to extend this stage by innovating, gathering customer insights, and exploring ways to maintain or increase their market share despite the fierce competition.**

emergence of substitutive product: orange or apple juice

Decline Stage

the opposite complement: coffee n sugar

During the decline stage, the product **experiences a drop in sales due to market saturation and the availability of alternative solutions**. Companies may reduce or cease marketing efforts if customer loyalty wanes in favor of competitors. Decisions are made whether to retire the product, discontinue support, or potentially revamp it entirely. Introducing a new or improved version can allow the company to re-enter the introduction stage of the product life cycle. An example of this is Microsoft's discontinuation of Windows 8.1 in 2023 to focus on newer technologies.

III. Tesla Motors¹

A. **Mission Statement:** Vision

The Future Is Autonomous self-organized

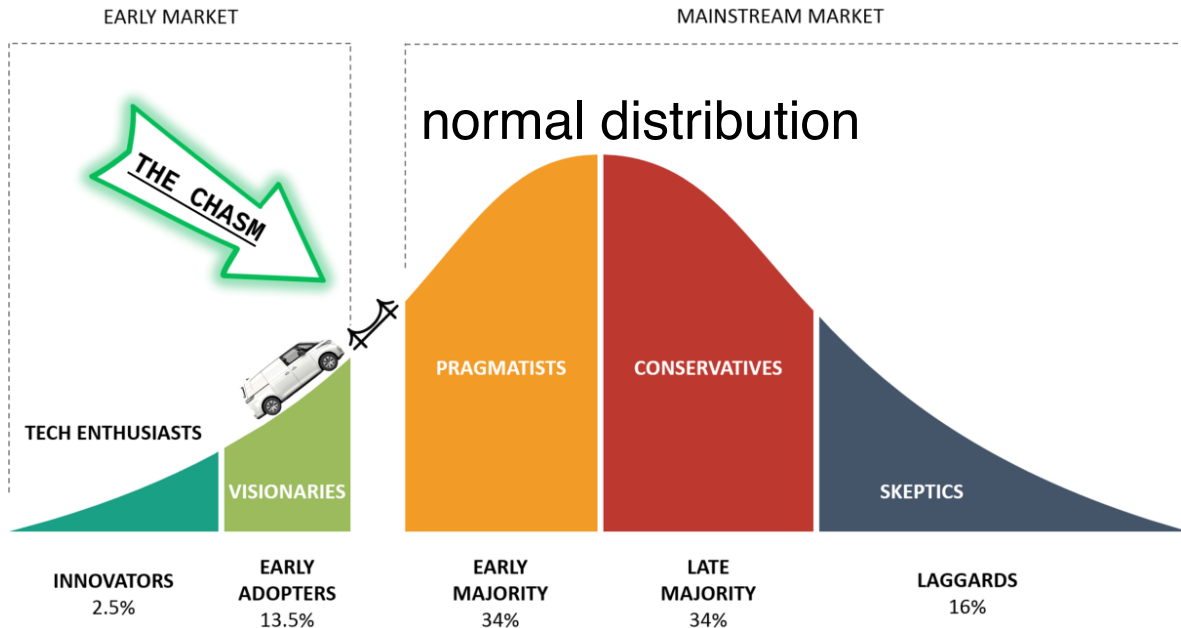
To create a **sustainable future**, we must democratize transportation. We do this by making driving more **efficient, affordable and safe**. Autonomy makes this future **possible, today.**²



¹Antti Lyyra and Kari Koskinen. "How Tesla is changing product life cycle in the car industry." LSE blogs. February 5th, 2018. <https://blogs.lse.ac.uk/management/2018/02/05/how-tesla-is-changing-product-life-cycle-in-the-car-industry/>.

² <https://www.tesla.com/we-robot>

B. Tesla's Product Life Cycle



It is rather difficult to locate the current status of Tesla onto the product life cycle. One potential solution is for us to look at the “Early Adopters Curve of product life cycle” which is a derived analytic tool in business. Based on the market targeted customer we can then infer the stage at which Tesla is undergoing backward. Most would argue that tesla’s development has just began its growth; some argues that it is at its maturity already. But only history knows better and there always lies the unpredictability in the world of commerce and economics.

Q2: What stage of development has Tesla reached for you? Justify your answer.